

Daily Research Digest — 2026-04-26

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

Summary of Key Research Papers in Astronomy and Astrophysics (23rd April 2026)

Here's a summary of some key research papers published on April 23, 2026: #### **Galaxies**

1. **Aspects of gravitational clustering and structure formation in the Universe**

- **Abstract:** This study explores how non-linear gravitational interactions influence galaxy distribution, halo abundance, and peculiar velocities using N-body simulations. It reveals deviations from theoretical predictions for the halo mass function (HMF), showing dependencies on the slope of the input power spectrum.
- **Citations (Semantic Scholar):** 0

2. **Multi-wavelength study of EP250416a / GRB 250416C: An Optically Dark Long GRB with a Late Jet Break**

- **Abstract:** This paper details the multi-wavelength study of the gamma-ray/X-ray transient EP250416a, observed by various telescopes. The burst has a duration of 30 seconds in X-rays and 17.7 seconds in gamma rays, with an isotropic energy of $E_{iso} = 2.7 \times 10^{50}$ erg (X-ray) and 7.34×10^{51} erg (gamma rays). The optical spectroscopy determined a redshift of $z = 0.963$.
- **Citations (Semantic Scholar):** 0

Dark Matter 1. **Saturation Mechanisms in the Interacting Dark Sector**

- **Abstract:** This research introduces phenomenological models with an interacting dark sector modulated by a sparseness scale, aiming to describe the late-time acceleration of the universe. The study examines phase-space dynamics and stability properties using Supernovae catalogues, cosmic chronometers, Baryon Acoustic Oscillations.
- **Citations (Semantic Scholar):** 0

2. **Conformal anomaly transport induced by dark photon**

- **Abstract:** This paper explores the effect of inhomogeneity in gravitational fields on transport effects predicted by a field theory describing massless Dirac fermions coupled to visible and hidden sectors via a dark photon model.

- **Citations (Semantic Scholar):** 0
3. **Constraining dark matter self-interaction from kinetic heating in neutron stars**
 - **Abstract:** This study investigates the role of dark matter self-interactions on neutron star temperatures, suggesting that observations by James Webb Space Telescope and future telescopes could provide constraints or signatures for dark matter self-interactions.
 - **Citations (Semantic Scholar):** 0
 4. **Saturation Mechanisms in the Interacting Dark Sector**
 - **Abstract:** This paper introduces a family of phenomenological cosmological models with an interacting dark sector modulated by a sparseness scale to explain late-time acceleration. The study examines phase-space dynamics and stability properties using Supernovae catalogues, cosmic chronometers, Baryon Acoustic Oscillations.
 - **Citations (Semantic Scholar):** 0
- ### **Dark Matter** 1. **Conformal anomaly transport induced by dark photon**
- **Abstract:** This research explores the impact of inhomogeneity in gravitational fields on transport effects predicted by a field theory involving massless Dirac fermions and hidden sectors via a dark photon model.
 - **Citations (Semantic Scholar):** 0
2. **Constraining dark matter self-interaction from kinetic heating in neutron stars**
 - **Abstract:** This paper investigates how dark matter self-interactions affect the temperatures of neutron stars, suggesting that future telescopes could provide stringent constraints or signatures for dark matter self-interactions.
 - **Citations (Semantic Scholar):** 0

These papers cover a range of topics including gravitational clustering in galaxies, properties of gamma-ray bursts, and various aspects of dark matter physics.

Paper Summaries and Figures

1. An RXTE Search for the Sterile Neutrino Decay in Galaxy Clusters

Focus: galaxy clusters, galaxies, dark matter

Authors: Mark J. Henriksen

Published: 2026-04-23T16:11:53+00:00

PDF: <https://arxiv.org/pdf/2604.21817v1>

Summary: We have searched for the 3.55 keV line from sterile neutrino decay using 3.1 megaseconds of RXTE cluster data. A 2.5σ excess of emission over a thermal model is found over the energy span of the 3.55 keV line in the combined spectra of the eight clusters that individually have an excess. The residuals are added to increase the signal to noise ratio of the excess, which is then modeled with a Gaussian to simulate the instrumental spectral response. We find a significant correlation ($r = 0.76$) for a line centered at 3.6 keV with a model flux of 3.07×10^{-5} ph cm $^{-2}$ s $^{-1}$. Mixing angle for detected clusters ranges from 0.35 to 6.2×10^{-10} . The decay rate inferred from the line flux is strongly correlated ($r = 0.87$) with cluster temperature, which is due to hotter, more massive clusters having a larger amount of dark matter. Approximately half of the decay line total flux comes from the Coma cluster. We fit the Coma cluster spectrum with two different three-component models. The first includes a Gaussian fixed at 3.55 keV to model soft emission. The second three-component model uses a second thermal component to model soft emission. The model fit parameters indicate that the second thermal component is modeling high-energy residuals rather than low ones, where the Gaussian is. Though our line fluxes exceed most reported detections and upper limits, they do not overproduce the dark matter. We conclude that some fraction of the marginally detected excess could be attributed to the decay line since low-temperature thermal emission and systematics fail to model it completely.

Selected figures

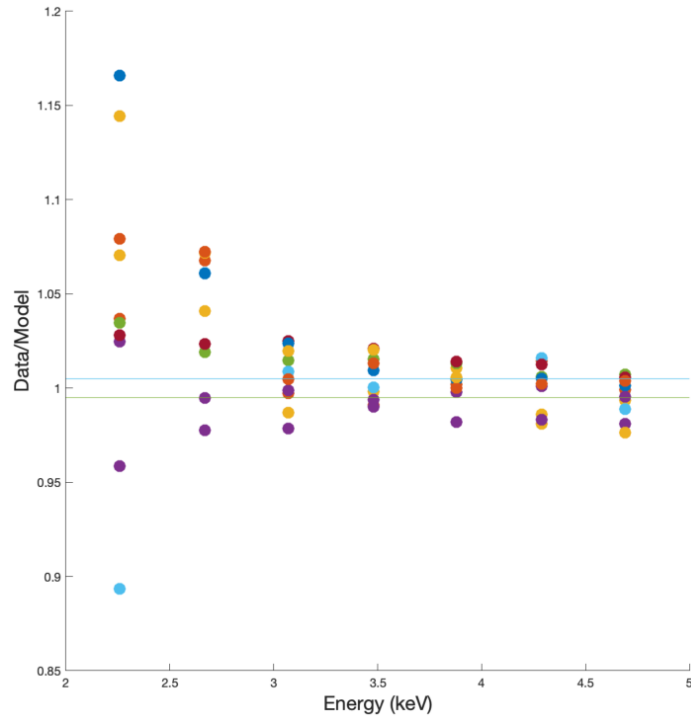


Figure 3. The ratio (data/model) in each channel covering the energy range of the 3.55 k feature folded through the RXTE PCA energy resolution is shown for each cluster. Lines that bracket the systematic error for the PCA as $\pm\sim 0.5\%$. The features can be compared systematic error and the number of c

2. Hamilton’s Object Revisited: A challenging source redshift of a strong lensing configuration

Focus: gravitational lensing

Authors: Jenny Wagner, Richard E. Griffiths, Emilio E. Falco

Published: 2026-04-21T18:00:01+00:00

PDF: <https://arxiv.org/pdf/2604.19867v1>

Summary: Low-resolution spectrographs used to have difficulties to determine redshifts of galaxies at $z \approx 1$ and $z \approx 3$. Spectral emission and absorption lines of magnesium and iron redshifted to $z \approx 1$ fall close to hydrogen, silicon, and oxygen lines at $z \approx 3$. Here, we demonstrate that, even with modern, integrated field unit spectrographs, this task remains challenging. Hamilton’s object, a blue star-forming galaxy, gravitationally lensed into three multiple images by the galaxy cluster SDSS J223010.47-081017.8 is such a case. Using the Blue Keck Cosmic Web Imager, its redshift was determined as $z = 0.82$, while its MOIRCS spectrum hinted at $z = 3.201$. To resolve the ambiguity, we completely re-analyse the Blue KCWI spectra of all three multiple images including the star-forming region in the outskirts. We employ a new data reduction pipeline, PypeIt, signal enhancement, and line fitting by Python-routines. The re-evaluation confirms the previous result based on 6 absorption features, $z = 0.820 \pm 0.001$ and 4 emission features, $z = 0.821 \pm 0.002$. The alternative $z = 3.199 \pm 0.003$, based on 6 absorption and 2 emission lines is a worse fit, also compared to other spectra. Moreover, we find the MOIRCS spectrum inconclusive: Observations cover two of three multiple images, with the slit for image C only covering its central bulge; furthermore the pixel-to-wavelength calibration requires a nightsky-emission-line calibration due to a missing calibration arc lamp. New MOIRCS observations are needed to verify that Hamilton’s object has the smallest separation in angular diameter distance between lensing cluster and source galaxy among the known cluster-scale strong lenses.

Selected figures

A&A proo

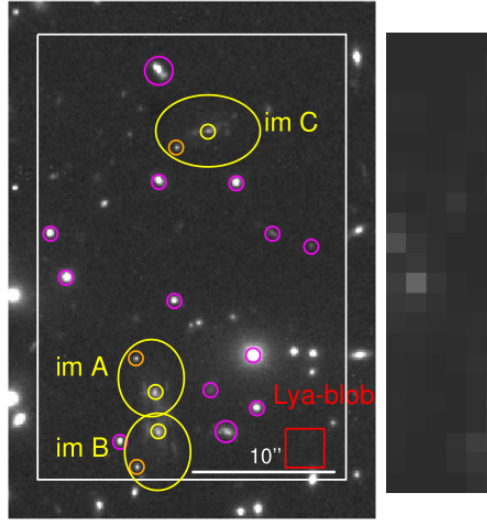


Fig. 2. Left: HST F110W filter-band observation with multiple images in the outer part of the galaxy are marked by yellow and orange circles. The field that may leave signatures in the white-light image of the KCWI observation (in the white rectangle) to match the resolution of the KCWI observation.

regions overlaid such that the regions coincide with the brightness in all pixels inside the yellow pixel masks (large ovals and small circles red box, discussed in Section 2.1. the KCWI data cube. To find optimum alignment, we mark some bright galaxies in the environment of the triple image co figure

3. Informative Priors on Primordial Non-Gaussianity Bias b_ϕ From Galaxy Formation

Focus: galaxies

Authors: Anne Moore, Lucia A. Perez, Elisabeth Krause

Published: 2026-04-23T15:46:53+00:00

PDF: <https://arxiv.org/pdf/2604.21790v1>

Summary: Constraining primordial non-Gaussianity via its scale-dependent imprint on galaxy clustering requires knowledge of the bias parameter b_ϕ , which is exactly degenerate with $f_{\text{NL}}^{\text{loc}}$ at leading order. To break this degeneracy, current analyses adopt the relation ($b_\phi = 2\delta_c(b_1 - 1)$) based on the assumption of a universal mass function. This relation is known to break down for physically motivated galaxy selections, introducing systematic errors in the inferred $f_{\text{NL}}^{\text{loc}}$ that scale directly with the assumed b_ϕ prior. We present a framework to construct physically motivated, observation-conditioned priors on b_ϕ by marginalizing over galaxy formation uncertainties. We use the CAMELS-SAM simulation suite, augmented by separate Universe simulations, to measure galaxy formation observables, like the stellar mass function (SMF) and the stellar-to-halo mass relationship (SHMR), and b_ϕ across a range of galaxy formation parameters. From these measurements, we construct a distribution of b_ϕ conditioned on observations, and we select our galaxy sample to resemble the DESI Emission Line Galaxy (ELG) sample. Conditioning on the SMF or SHMR decreases σ_{b_ϕ} from 0.69 to 0.08 and 0.02 respectively – reductions of 88% and 97% – with consistent results when conditioning on the observed data directly. Despite substantial shifts in the galaxy formation posteriors driven by known SC-SAM discrepancies at high halo masses, the resulting b_ϕ distributions remain mutually consistent across all observables. The SMF and SHMR are found to carry sufficient constraining power to reduce the galaxy formation uncertainty in b_ϕ relevant for $f_{\text{NL}}^{\text{loc}}$ inference with next-generation spectroscopic surveys

4. Photometric Super-Resolution for Improving Galaxy Morphological Measurements using Conditional Generative Adversarial Networks

Focus: gravitational lensing

Authors: Samuel Kahn, Ryan Hausen, Hubert Bretonnière, Nicole Drakos, Brant Robertson

Published: 2026-04-22T05:23:02+00:00

PDF: <https://arxiv.org/pdf/2604.20195v1>

Summary: The measurement of galaxy morphological parameters from astronomical images features in a wide range of modern analyses, including galaxy evolution and cosmological weak lensing studies. The precision and accuracy of morphological parameter estimation can be influenced by several key factors. The effective seeing of the image, summarized by the point spread function (PSF), limits how galaxy features or light profiles are resolved. The pixel scale of the detector also influences the resolution and the amount of statistical information available for a given object. The depth of the observations determines the signal-to-noise ratio of the image. Improving each of these factors is very costly, either in terms of detector upgrades, observatory design, or observing time. Here, we develop a conditional generative adversarial network, called Neo, trained to transform existing ground-based images into sharper, finer-scale images comparable to space-based image quality. We demonstrate that Neo improves the accuracy of measured morphological parameters by factors of 2-10 when trained to translate Subaru Hyper Suprime-Camera (HSC) images to approximate Hubble Space Telescope (HST) data. Neo is designed for applicability to ongoing, large-scale surveys such as the Legacy Survey of Space and Time (LSST) conducted by Vera C. Rubin Observatory in combination with space telescopes such as HST, James Webb Space Telescope, and Nancy Grace Roman Space Telescope. These results suggest that Neo could be used to improve both cosmological and galaxy evolution analyses based on massive, ground-based survey datasets like LSST. The model code is open source and available at <https://purl.archive.org/neo/code>.

Selected figures

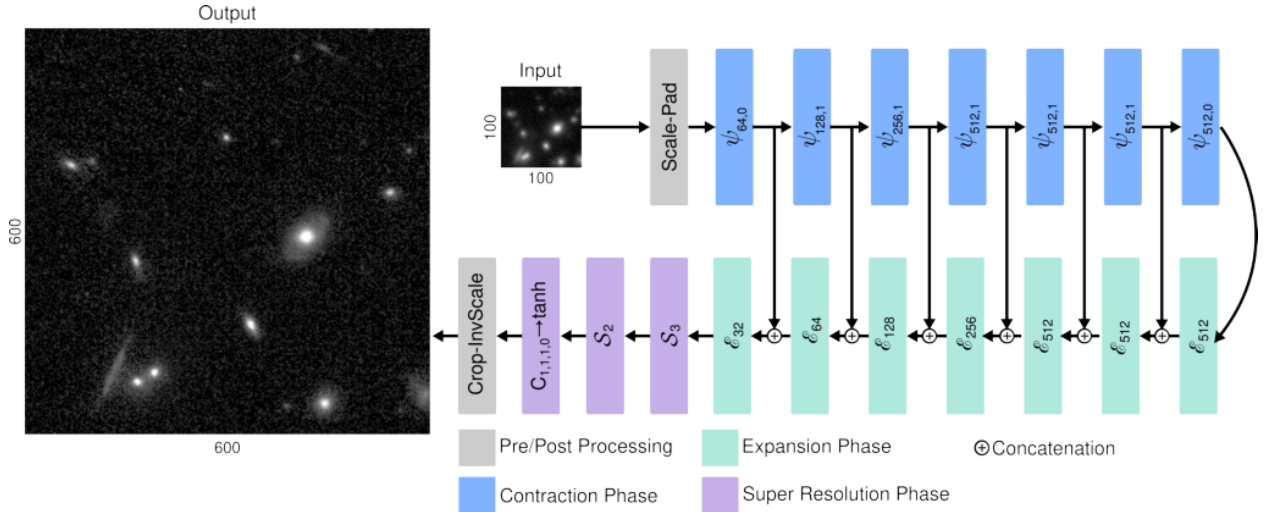


Figure 1. Schematic of the Neo generator network including pre-processing and post-processing steps. In the Neo neural network, the generator is defined by a contraction phase, expansion phase, and super-resolution phase. The contraction operators $\psi_{sif,b}$ reduce the spatial extent of the tensors by a factor f .

5. Precision Kinematic Sunyaev–Zel’dovich Measurements Across Halo Mass and Redshift with DESI DR2 and ACT DR6: Part II. Bright Galaxy Survey and Emission-Line Galaxies

Focus: gravitational lensing

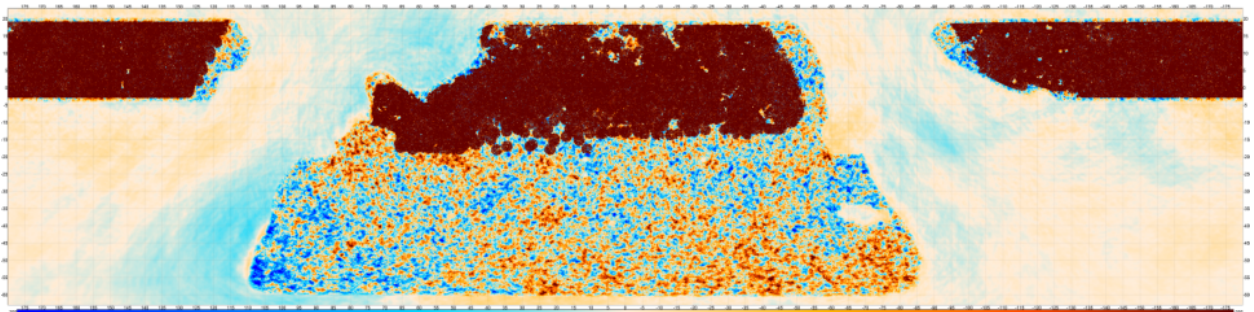
Authors: B. Hadzhiyska, S. Ferraro, F. J. Qu, B. Ried Guachalla, E. Schaan, J. Aguilar, S. Ahlen, D. Bianchi, D. Brooks, F. J. Castander, E. Chaussidon, T. Claybaugh, A. de la Macorra, Arjun Dey, Biprateep Dey, P. Doel, J. E. Forero-Romero, E. Gaztañaga, S. Gontcho A Gontcho, G. Gutierrez, J. Guy, K. Honscheid, C. Howlett, D. Huterer, M. Ishak, R. Joyce, R. Kehoe, T. Kisner, A. Kremin, O. Lahav, M. Landriau, L. Le Guillou, A. Leauthaud, M. Manera, P. Martini, A. Meisner, R. Miquel, S. Nadathur, N. Palanque-Delabrouille, W. J. Percival, F. Prada, I. Pérez-Ràfols, G. Rossi, L. Samushia, E. Sanchez, E. F. Schlafly, D. Schlegel, J. Silber, D. Sprayberry, G. Tarlé, B. A. Weaver, R. Zhou, H. Zou

Published: 2026-04-21T17:59:30+00:00

PDF: <https://arxiv.org/pdf/2604.19745v1>

Summary: We present the first high-significance spectroscopic stacked kinetic Sunyaev-Zel’dovich (kSZ) measurements of circumgalactic gas profiles for both Bright Galaxy Survey (BGS) and Emission Line Galaxy (ELG) tracers, combining DESI Data Release 2 with ACT Data Release 6. Using reconstructed line-of-sight velocities from the DESI galaxies and high-resolution ACT temperature maps, we detect the kSZ signal at high significance, reaching signal-to-noise ratios of up to ~ 9 for BGS and ~ 7.5 for ELGs in optimal stellar-mass selections. Together with the LRG measurements presented in Paper I, these constitute the most significant kSZ detections from any spectroscopic survey to date. We perform the analysis in both real and harmonic space, obtaining consistent results. By splitting both tracers into stellar-mass bins, we study the scaling of the kSZ amplitude with galaxy properties. Combining the kSZ measurements with ACT Data Release 6 (DR6) CMB lensing maps enables a joint calibration of the galaxy-halo connection and the gas fractions of host halos. For the BGS galaxies, we observe low gas fractions around the virial radius relative to standard expectations, likely attributable to active galactic nuclei (AGN) activity. We find some evidence for higher-mass halos retaining a larger fraction of their baryons, consistent with more efficient feedback in lower-mass systems. For the ELG sample, dominated by blue, star-forming galaxies, we provide the first detection of the gas distribution in ELG host halos. The ELGs appear to exhibit relatively high gas fractions, which points to the possibility of weaker feedback (due to e.g. low AGN and supernova feedback activity) at their mass scale. Finally, we present generalized Navarro-Frenk-White (GNFW) fits to the harmonic-space measurements, providing a compact parametrization of gas profiles for forward modeling in large-scale structure analyses.

Selected figures



6. SDSSJ110546.07+145202.4: The first long-duration radio changing-look NLS1 galaxy

Focus: gravitational lensing

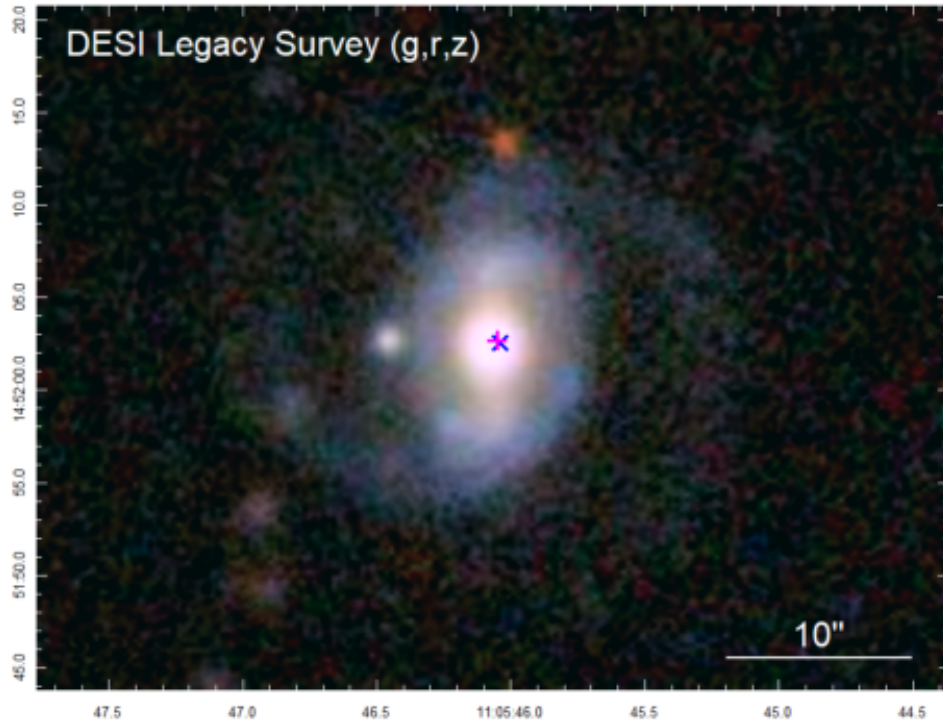
Authors: S. Komossa, D. Grupe, A. Kraus, P. G. Edwards, E. F. Kerrison, K. Rose, R. Soria, T. An, M. J. Hardcastle, K. E. Gabanyi, S. Panda, D. W. Xu, J. Wang, S. Frey, A. Mezosi

Published: 2026-04-21T13:09:40+00:00

PDF: <https://arxiv.org/pdf/2604.19435v1>

Summary: SDSSJ110546.07+145202.4 stands out as a unique radio changing-look Narrow-line Seyfert 1 (NLS1) galaxy that has brightened dramatically and shows an exceptionally long duration of its "on" phase. We present the first high-frequency radio observations, the first simultaneous radio spectral energy distributions (SEDs), the first optical–UV–X-ray SEDs, and the first X-ray monitoring and spectroscopy of this recently discovered event. Importantly for understanding the nature of the outburst, we show that the X-ray spectrum is soft with a photon index $\Gamma_X=2.5$; line-of-sight absorption and extinction are low or absent; the radio SED is peaked at low frequencies ~ 2 GHz; and the radio outburst emission is very long-lived ($t > 8$ yr) and roughly constant. The softness of the X-ray spectrum, low supermassive black hole (SMBH) mass, and high Eddington ratio all corroborate the optical NLS1 classification. We discuss multiple outburst scenarios, including lensing, absorption, a binary SMBH merger, a long-duration giant-star tidal disruption, a newly ignited active galactic nucleus (AGN), and an accretion-rate change. While most of them can be either excluded or are deemed too rare and lack positive evidence so far, most or all types of these transients are expected to be detected in ongoing VLA and upcoming SKA surveys. SDSSJ110546.07+145202.4 itself is well explained by an accretion rate change that triggered the powerful radio jet emission. The low redshift and SMBH mass of this system offer a unique perspective of the physical processes of radio-jet ignition that are expected to operate in the early Universe around growing SMBHs.

Selected figures



6. SMBH MASS ESTIMATES, EDDINGTON RATIO AND NLS1 CLASSIFICATION

7. Accelerating scaling solutions from dark matter particle creation

Focus: dark matter

Authors: Sudip Halder, Jaume de Haro, Supriya Pan, Emmanuel N. Saridakis, Tapan Saha, Subenoy Chakraborty

Published: 2026-04-23T15:28:34+00:00

PDF: <https://arxiv.org/pdf/2604.21770v1>

Summary: This article opens new window to obtain accelerating scaling attractors without any need of dark energy. We study cosmological dynamics in a two-fluid system where pressureless dark matter (DM) undergoes adiabatic particle creation and exchanges energy with a barotropic fluid. Considering six widely used interaction prescriptions, we formulate the corresponding autonomous systems in a compact phase space and perform a unified dynamical analysis. We find that accelerating scaling attractors, namely late-time states where both fluids coexist with fixed energy fractions, arise only when the interaction is controlled by the DM density and energy flows from DM to the second fluid. Such attractors appear in the global and local DM-based interactions, and in the global mixed case, but are entirely absent when the interaction depends on the second fluid or on local mixed terms, which instead drive the universe to a DM-dominated accelerating phase. These results clarify the unique conditions under which matter creation can mimic dark-energy-like behaviour without introducing a dark-energy component.

Selected figures

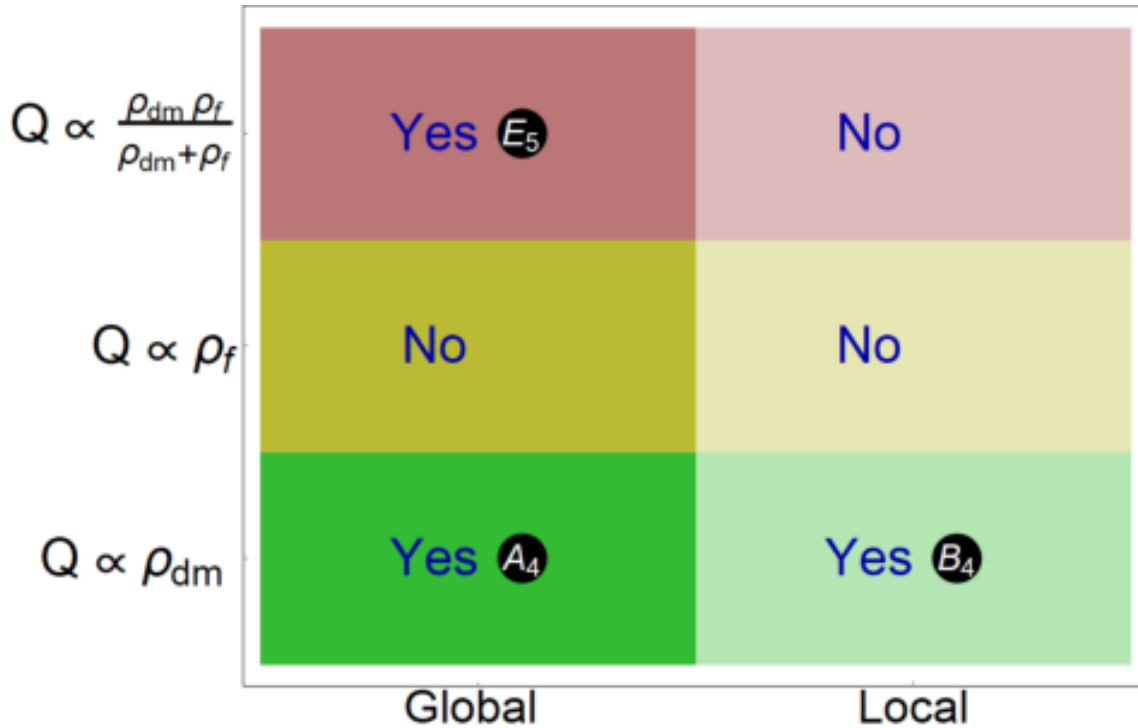


FIG. 7. Summary of accelerating scaling attractors identified in this study. Each row corresponds to a particular energy density (green, yellow, or pink), while each column represents the type of interaction: global rate (dark shading) or local rate (light shading).

8. Delving into the depths of NGC 3783 with XRISM: V. Broad-band modeling of ionized outflows

Focus: galaxies

Authors: Keqin Zhao, Jelle S. Kaastra, Liyi Gu, Missagh Mehdipour, Megan E. Eckart, Keigo Fukumura, Matteo Guainazzi, Chen Li, Christos Panagiotou, Matilde Signorini

Published: 2026-04-23T14:16:59+00:00

PDF: <https://arxiv.org/pdf/2604.21710v1>

Summary: The Seyfert 1 galaxy NGC 3783 hosts a multiphase warm absorber (WA) that has been extensively studied in the X-ray band. High-resolution spectra from 2000-2001 revealed a complex outflow with multiple ionization and velocity components. Two decades later, new XMM-Newton and XRISM observations allow us to investigate the long-term evolution of these outflows. We perform joint spectral modeling of the XMM-Newton/RGS and XRISM/Resolve time-averaged spectra using the pion photoionization code within SPEX. We derive the ionization parameter, column density, turbulent velocity, and outflow velocity for each absorption component, and investigate their thermal stability and Absorption Measure Distribution (AMD) to characterize the physical and dynamical properties of the WA in NGC 3783 in 2024. We compare these results with the 2000-2001 epoch to assess long-term variability, stability, and possible changes in the absorber population. We identify eight WA components spanning $\log \xi = 1.08\text{--}3.38$ and outflow velocities of $480\text{--}1230 \text{ km s}^{-1}$. The ranges of column densities and turbulent velocities remain broadly consistent with the WAs from 2000-2001, but the earlier data contained more low-ionization, high-velocity components. The total column density in 2024 is 1.5 times larger than in 2000-2001, requiring replenishment by fresh material. The dominant Unresolved Transition Array (UTA) absorber (Comp. B3) has increased its column density by a factor of three while maintaining a similar ionization parameter. The WAs in NGC 3783 have undergone significant structural and dynamical evolution over the past 24 years.

Selected figures

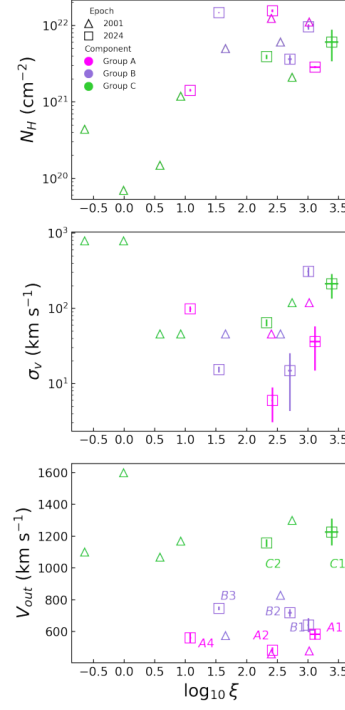


Fig. 3. Relations between the parameters of the eight outflow components derived from the XRISM Resolve spectrum of NGC 3783. The top panel displays the column density N_H as functions of the ionization parameter ($\log \xi$). The middle and bottom panels show the turbulent velocity (σ_v) and outflow velocity (v_{out}) as functions of $\log \xi$.

9. Saturation Mechanisms in the Interacting Dark Sector

Focus: dark matter

Authors: Andronikos Paliathanasis, Kevin J. Duffy

Published: 2026-04-23T13:36:31+00:00

PDF: <https://arxiv.org/pdf/2604.21671v1>

Summary: We introduce a family of phenomenological cosmological models featuring an interacting dark sector modulated by a sparseness scale parameter, in order to describe the late-time accelerated expansion of the universe. The sparseness scale, inspired by well-established saturation mechanisms in ecology and biology, is introduced in the interaction as a half-saturation constant that bounds the energy exchange between dark matter and dark energy, controls the dynamical behaviour of the physical variables and can prevent the phantom crossing. We consider three nonlinear interacting models, where two of them recover the linear interacting scenarios when the sparsity parameter vanishes. We examine the phase-space of the cosmological field equations by using the Hubble normalization approach. We determine the stationary points and their stability properties in order to reconstruct the asymptotics behaviour of the field equations. Such an analysis allows us to demonstrate the effects of the sparseness scale on the background dynamics. We test the interacting models with observational data. Specifically, we employ Supernovae catalogues, cosmic chronometers, Baryon Acoustic Oscillation measurements from DESI DR2, and redshift-space distortion measurements of the growth of large-scale structure through the f and $f\sigma_8$ observables. The Bayesian analysis suggests that, for two of the three models, a vanishing sparsity parameter is disfavoured at more than the 95\% confidence interval, providing observational support for a nonzero sparseness scale in the dark sector interaction.

10. Conformal anomaly transport induced by dark photon

Focus: dark matter

Authors: Marek Rogatko, Karol I. Wysokinski

Published: 2026-04-23T13:27:00+00:00

PDF: <https://arxiv.org/pdf/2604.21664v1>

Summary: We have considered the problem of the influence of inhomogeneity of gravitational field on transport effects predicted by the field theory describing massless Dirac fermions in the Maxwell and dark matter background. As a model of dark sector one takes into account dark photon model, where the hidden sector is described by the auxiliary U(1)-gauge field coupled to the visible sector. Elaborating the model we restrict our considerations to the case when Weyl type conformal transformation slightly differs from the Minkowski spacetime. This assumption simplifies the calculations and enables us not to use complicated methods of the quantum field theory in the curved background. The resulting currents stemming both from visible and dark sectors are proportional to the adequate beta functions appearing in the elaborated systems. For charge-less dark sector we predict corrections to the scale conductivities in both sectors: linear in α in the dark sector and quadratic in the visible one.

11. Constraining dark matter self-interaction from kinetic heating in neutron stars

Focus: dark matter

Authors: Sambo Sarkar

Published: 2026-04-23T13:17:20+00:00

PDF: <https://arxiv.org/pdf/2604.21652v1>

Summary: Dark matter search strategies have started advancing towards the neutrino fog. In this regard, compact objects such as neutron stars have already demonstrated their ability in probing such low DM-nucleon cross-sections from dark matter induced effects. In the optically thin limit, effect of dark matter self-interaction becomes relevant and may assist the capture and thermalization of dark matter inside stars, imparting observable changes on neutron star temperatures. The resulting radiation although weak can be potentially detected by the James Webb Space Telescope and upcoming Thirty Meter Telescope and the European Extremely Large Telescope. Observation of cold neutron stars accompanied by advancements in direct detection probes would provide stringent constraints or a smoking-gun signature for dark matter self-interactions. The potential detection of a neutron star with surface temperatures $\sim (1000 - 1200)$ K in the optically thin limit can push the bounds on asymmetric dark matter self-interaction cross-section to approximately two orders of magnitude more stringent than the bullet cluster.

12. Aspects of gravitational clustering and structure formation in the Universe

Focus: galaxies

Authors: Swati Gavas

Published: 2026-04-23T12:52:25+00:00

PDF: <https://arxiv.org/pdf/2604.21634v1>

Summary: The distribution of galaxies, halo abundance, and peculiar velocities are influenced by non-linear gravitational interactions, making the study of non-linear evolution crucial for accurate cosmological predictions. We explore these aspects using N-body simulations. Theoretical models of the halo mass function (HMF) can be formulated without referencing a cosmological model or input power spectrum. HMF obtained from N-body simulations show systematic deviations of 5-20\% from theoretical predictions. The physical origin of deviations may result from cosmology, the power spectrum, or both. We examine HMF deviations from universality for scale-free power spectra with an Einstein-de Sitter cosmology. We demonstrate that the mass function exhibits an explicit dependence on the slope of the input power spectrum. We find that an effective index of the Λ CDM model can correspond to the HMF from scale-free cosmologies as a first approximation. Furthermore, structure formation has led to deviations from homogeneity and isotropy on scales up to at least 100 Mpc/h, expected to affect measurements of H_0 . We revisit this issue of the concordance model. We find a correlation between errors in H_0 estimates and the density around the observer. Further, our mock observations reveal that deviations of up to 5\% can occur in Milky Way-sized halos. While this finding alone does not fully resolve the Hubble tension, it may account for part of it. It is essential to understand the limitations of N-body simulations to avoid misinterpreting data. We show that the missing power at small scales introduces errors in the root-mean-square fluctuations and in the simulated mass function. Our analytical calculation indicates that mode coupling between small and large scales depends on resolving collapsed halos. Therefore, accurate mode coupling estimates require sufficient halos in the simulation.

13. Multi-wavelength study of EP250416a / GRB 250416C: An Optically Dark Long GRB with a Late Jet Break

Focus: galaxies

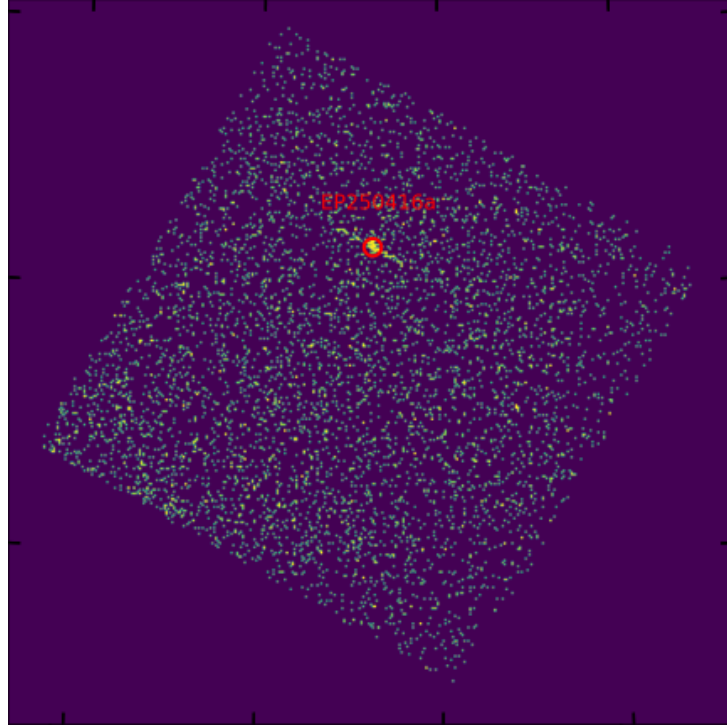
Authors: Guoying Zhao, Duo-Le Cao, Rong-Feng Shen, Hui Sun, Chi-Chuan Jin, Wei-Min Yuan, Chen-Wei Wang, Shao-Lin Xiong, Dmitry Svinkin, Dong Xu, Shuai-Qing Jiang, Peter G. Jonker, Yun Wang, Hao Zhou, Chang Zhou, Xinlei Chen, Kaushik Chatterjee, Xue-Feng Wu, Xiao-Feng Wang, Chun Chen, Yuan Liu, Andrew J. Levan, Jennifer Alexandra Chacon Chavez, Jonathan Quirola-Vásquez, Franz E. Bauer, Antonio Martin-Carrillo, Gregory Corcoran, Daniele B. Malesani, Dmitry Frederiks, Anna Ridnaia, Alexandra L. Lysenko, Mikhail Ulanov

Published: 2026-04-23T12:44:15+00:00

PDF: <https://arxiv.org/pdf/2604.21624v1>

Summary: We present multi-wavelength study of the γ /X-ray transient EP250416a (also designated GRB 250416C), triggered by the Einstein Probe (EP) Wide-field X-ray Telescope and also by SVOM and Konus-Wind. Observations spanning the gamma-ray, X-ray, and optical bands facilitated detailed analysis of the burst’s prompt emission, afterglow evolution, and physical origin. EP250416a exhibits a burst duration of 30 s in X-ray and 17.7 s in gamma-rays, with joint spectral fitting of 0.5–5000 keV data gives $E_{\text{peak}} = 342^{+90}_{-232}$ keV. Optical spectroscopy of the afterglow, acquired with the Gemini Multi-Object Spectrograph (GMOS) on Gemini South, yielded a redshift of $z = 0.963$. Accounting for the measured redshift, the isotropic energies are $E_{\text{X,iso}} = 2.7^{+0.9}_{-0.5} \times 10^{50}$ erg and $E_{\gamma,\text{iso}} = 7.34^{+5.1}_{-2.1} \times 10^{51}$ erg, aligning with the Amati relation for long GRBs. The fluence ratio $S(25 - 50 \text{ keV})/S(50 - 100 \text{ keV}) = 0.78^{+0.1}_{-0.15}$ classifies EP250416a as an X-ray rich (XRR) GRB. The X-ray afterglow shows an initial shallow decay ($\alpha \approx -0.5$) transitioning to a canonical decay phase ($\alpha \approx -1$), with a very late jet break at $t \sim 1.5 \times 10^6$ s, corresponding to a jet half-opening angle of $\theta_j = 10.6^{+1.9}_{-1.8}$ degrees. EP250416a is optically dark, as it shows only a faint r -band detection ($r = 24.16$ mag) from Gemini South-GMOS and a low optical-to-X-ray spectral index $\beta_{\text{OX}} = 0.3$. This may be attributed to significant host-galaxy extinction, with a required $A_V^{\text{host}} = 5.5$ mag derived from the extinction curve model.

Selected figures



mage of the EP250416a captured by the EP/WXT CMOS1 chip at 17:53:59 UTC on 16 April 2025. the source is localize 256.42and Dec. = 25.78. tions of EP250416a : the first at ~ 4.8 hours (PI: Kennea scnd at ~ 20.5 hours (PI: EPSC) after the WXT trigge bservations complemented the EP/FXT data by coverin